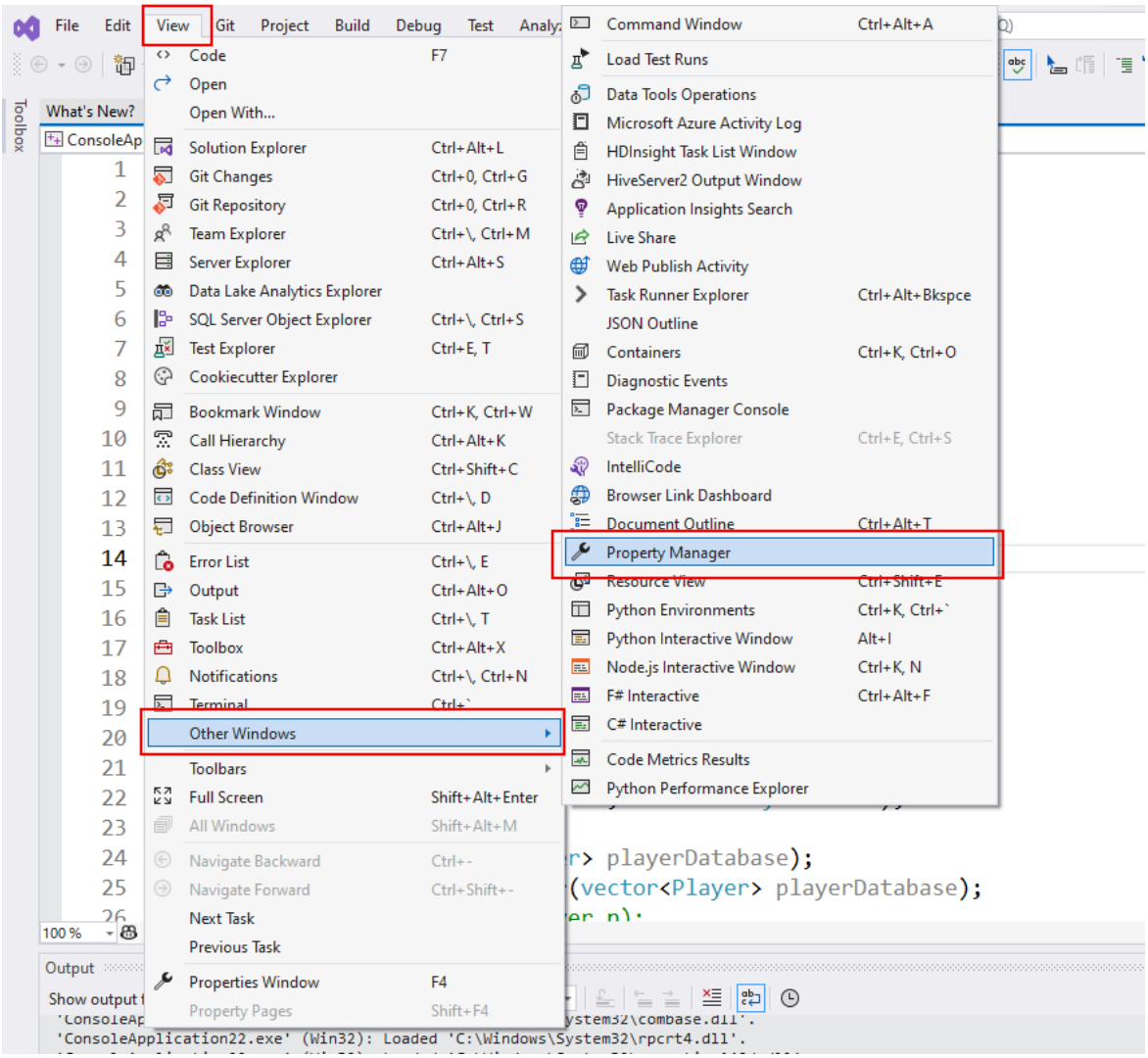
Using the C++ Boost Library inside Visual Studio

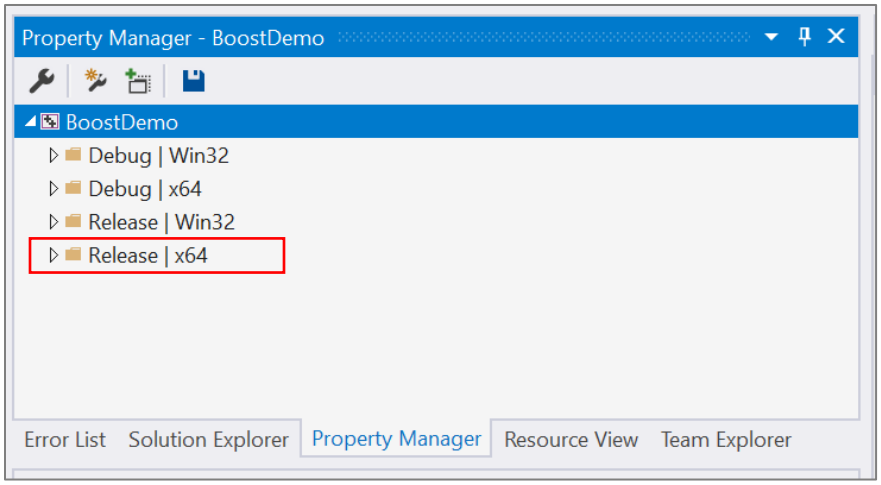
Preliminary Steps – Installing Boost, Adding a Property Sheet to a C++ Project

1. Visit the Boost website https://www.boost.org/users/download/
   Download the most current version of boost and install it in your machine (all you need to do is to unzip the file and place the folder is a designated area). In our example it was deployed in the **c:\Boost** folder as **c:\Boost\boost_1_69_0**

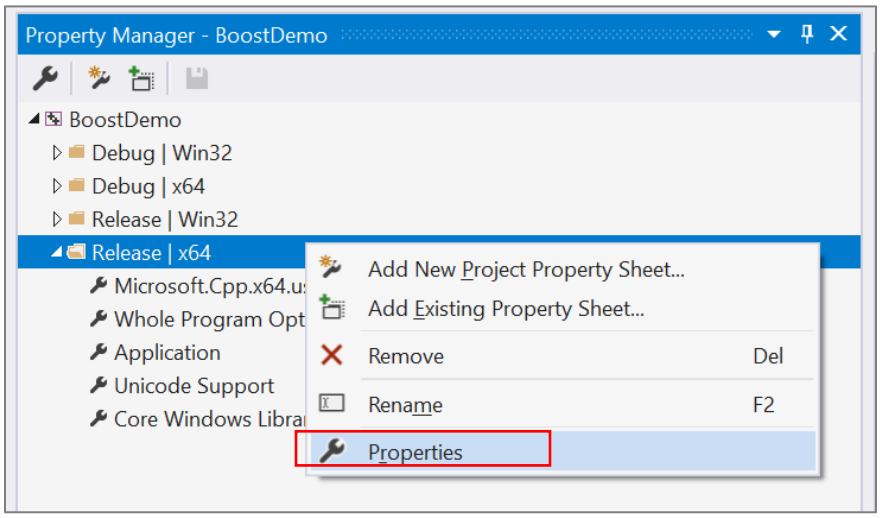
2. Create a VS C++ project. Call it **BoostDemo**.

3. Click on the *Property Manager* tab. If it is not already available on the UI, click on the menu **View > Other Screens > Property Manager**.



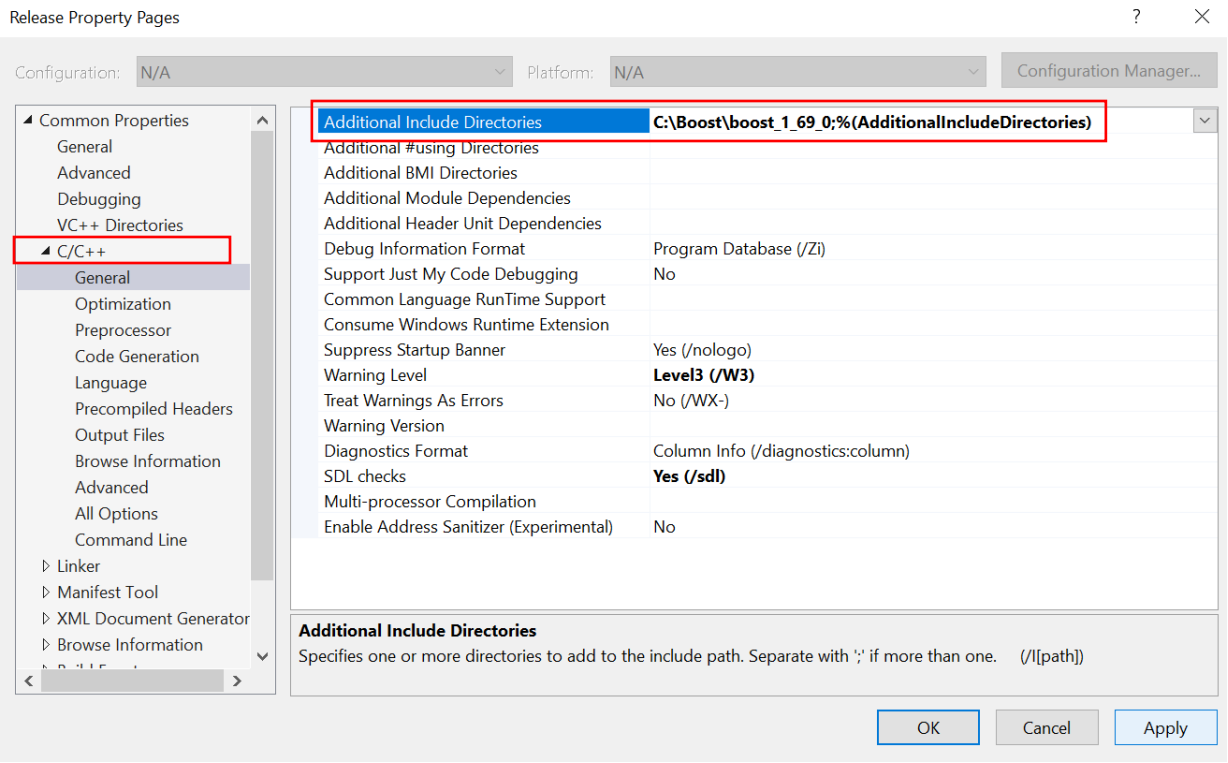


4. We have chosen to modify an existing Property Sheet (Expand the entry **Release | x64**). Right-click on the entry **Release | x64**. From the menu, select the option **Properties**. Click it.

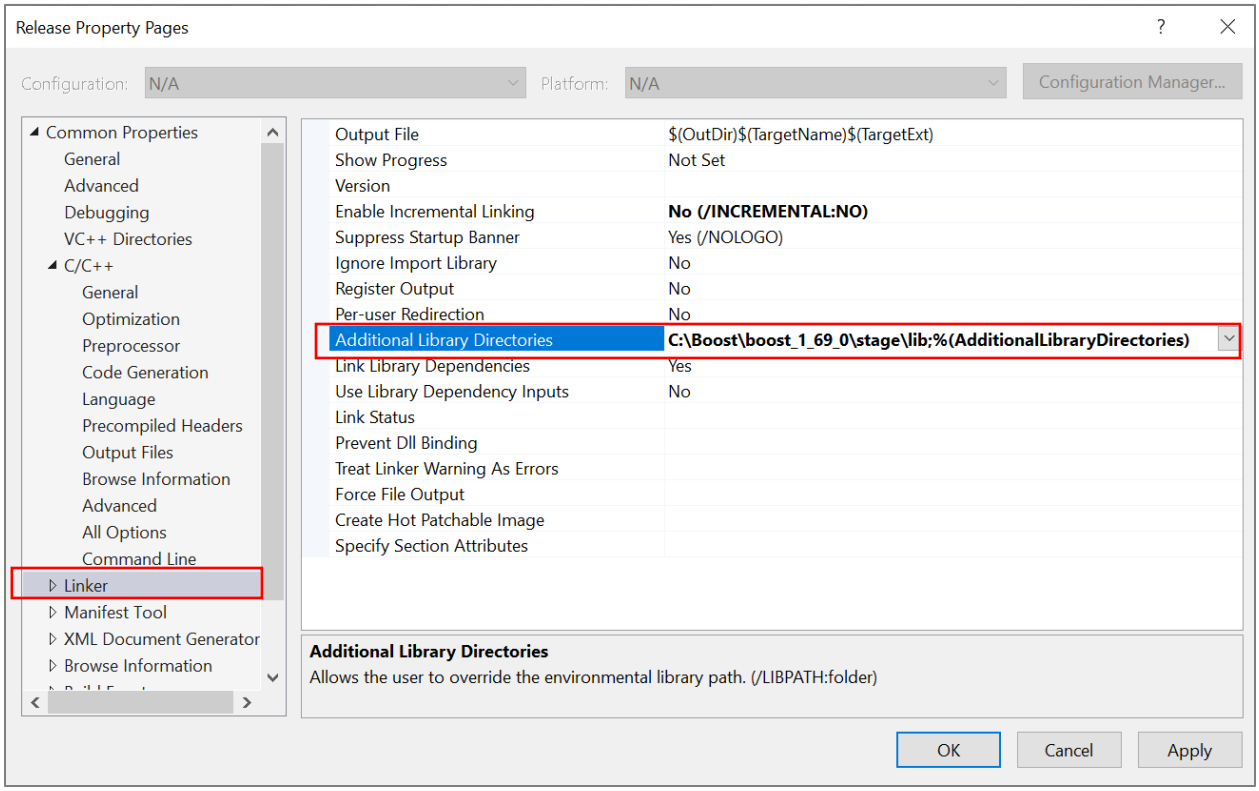


5. On the new page expand the **C/C++** entry. Modify the field **Additional Include Directories**. Enter the location of your Boost library. In this example **c:\Boost\boost_1_69_0**
Click **Apply** button.

[Alternatively, you may use the UI interface to locate and select the folder's path. Click on the *Drop-down arrow, <Edit…>, Plus-Folder icon*, pick location from *File Chooser*, click *the Select Folder* button, click *OK*.]



6. On the above Property Page click on the **Linker** option.
   Modify the entry **Additional Library Directories**. Enter the path **c:\Boost\boost_1_69_0\stage\lib**



7. Click the **OK** button. You are done!

Testing the library – Working with BIG numbers

The following app uses the Boost's Multiprecision library to work with large numbers. The app will

(1) calculate the factorial of a large number (say 40), and

(2) calculate the value of Euler's number *e* (e = $\sum_{n=0}^{\infty} \left(\frac{1}{n!}\right)$)

```
#include <boost/multiprecision/cpp_int.hpp>
#include <boost/multiprecision/cpp_dec_float.hpp>
#include <boost/math/constants/constants.hpp>

using namespace boost::multiprecision;
using namespace std;
//-------------------------------------------------------------------------------
cpp_dec_float_50 inverseFact(cpp_dec_float_50 n)
{
    cpp_dec_float_50 f = 1.0;
    for (cpp_dec_float_50 i = 1; i <= n; i++) {
        f *= i;
    }
    return 1 / f;
}
//-------------------------------------------------------------------------------
cpp_int boostFactorial(cpp_int num)
```

```
{
    cpp_int fact = 1;
    for (cpp_int i = num; i > 1; --i)
        fact *= i;
    return fact;
}
//-------------------------------------------------------------------------------------
int main()
{
    //calculate 40! (It will fail with a C++ long long type)
    //working with large integers
    cpp_int num = 40;
    cpp_int result = boostFactorial(num);

    cout << "\nFactorial of " << num << " is \n" << result << endl;

    //working with large precision decimal numbers
    cpp_dec_float_50 inverse = cpp_dec_float_50(1.0) / cpp_dec_float_50(result);

    cout << setprecision(numeric_limits<cpp_dec_float_50>::digits10);
    cout << "\nInverse of factorial(" << num << ") is \n" << inverse << endl;

    //Calculating an approximation of Euler's number
    cout << "\nEuler's Number\n";
    cpp_dec_float_50 sum = 0.0;
    for (cpp_dec_float_50 i = 0; i < 40; i++)
    {
        sum += inverseFact(i);
    }
    cout << "e = Sum(1/1!..1/40!) is  \n" << sum << endl;
}
```

Output

```
Factorial of 40 is

815915283247897734345611269596115894272000000000


Inverse of factorial(40) is

1.2256174391283858494235399849397569237255619644783e-48


Euler's Number

e = Sum(1/1!..1/40!) is

2.7182818284590452353602874713526624977572470936987
```

APPENDIX

Extracted from https://www.boost.org/doc/libs/1_72_0/libs/multiprecision/doc/html/boost_multiprecision/intro.html

Boost– Multiprecision Library

- The Multiprecision Library provides **integer, rational, floating-point**, and **complex** types in C++ that have more range and precision than C++'s ordinary built-in types.
- The Multiprecision types can interoperate with any fundamental (built-in) type in C++ using defined conversion rules.
- Depending on the selected type, precision may be arbitrarily large or fixed (for example, 50 or 100 decimal digits).
- There are several providers (called backends) such as GMP, MPFR, MPIR, MPC, and TomMath. These suppliers offer different capabilities. We will use plain Boost libraries.

Basic Data Types

The fundamental number types are

Type	Used for	Header
`cpp_int`	Multiprecision integers	`boost/multiprecision/cpp_int.hpp`
`cpp_rational`	Rational types	`boost/multiprecision/cpp_int.hpp`
`cpp_bin_float`	Floating-point types	`boost/multiprecision/cpp_bin_float.hpp`
`cpp_dec_float`	Floating-point types	`boost/multiprecision/cpp_dec_float.hpp`
`cpp_complex`	Complex types	`boost/multiprecision/cpp_complex.hpp`

For example, if you want a signed, 128-bit fixed size integer:

```
#include <boost/multiprecision/cpp_int.hpp>  //  Integer types.

boost::multiprecision::int128_t  my_128_bit_int;
```

Some programmers use **`typedef`** to create more convenient shorter names. For instance, you may use the following clause

```
typedef boost::multiprecision::int128_t  bigintT;

bigintT myIntVar;

myIntVar = 12345678901234567890;

cout << (myIntVar + 1) << endl;
```

Another very common coding style that reduces typing is shown below

```
namespace mp = boost::multiprecision;     // Reduces typing

mp::int128_t a(3), b(4);        // 128-bit integers
mp::int512_t c(50), d;          // 512-bit integers

b = 25;

d = c * a;                      // OK, result is an int512_t

cout << (2 * d + 1) << endl;    // expression uses common arithmetical ops and big integers
```

The following code fragment illustrates how to obtain the max value that different multiprecision cpp_int types have.

```
cpp_int limit1 = numeric_limits<int128_t>::max();
cout << "LIMIT128 \n" << limit1 << endl;
cout << "Has " << limit1.str().size() << " digits\n";

cpp_int limit2 = numeric_limits<int256_t>::max();
cout << "LIMIT256 \n" << limit2 << endl;
cout << "Has " << limit2.str().size() << " digits\n";

cpp_int limit3 = numeric_limits<int512_t>::max();
cout << "LIMIT512 \n" << limit3 << endl;
cout << "Has " << limit3.str().size() << " digits\n";

cpp_int limit4 = numeric_limits<int1024_t>::max();
cout << "LIMIT1024 \n" << limit4 << endl;
cout << "Has " << limit4.str().size() << " digits\n";
```

```
LIMIT int128_t
340282366920938463463374607431768211455
Has 39 digits

LIMIT int256_t
115792089237316195423570985008687907853269984665640564039457584007913129639935
Has 78 digits

LIMIT int512_t
13407807929942597099574024998205846127479365820592393377723561443721764030073546976801874298166903427690031858186486050853753882811946569946433649006084095
Has 155 digits

LIMIT int1024_t
179769313486231590772930519078902473361797697894230657273430081157732675805500963132708477322407536021120113879871393357658789768814416622492847430639474124377767893424865485276302219601246094119453082952085005768838150682342462881473913110540827237163350510684586298239947245938479716304835356329624224137215
Has 309 digits
```